CUMULATIVE NAVIGATION / +C, +C2 AND +C3

Quick reference card

Select a pad: **INFO** opens its waveform; **INFO** again opens parameters. From Menu, open **Tools** for globals. Labels identify the first custom release: +c (c1), +c2 or +c3.

Waveform [+c]	Vertical two-finger spread/pinch: amplitude **25-3200%**. Horizontal: time. Stationary two-finger tap: **100% + horizontal fit**.
Dial reset [+c]	Double-tap a numeric dial to restore its base value. Modulation stays assigned.
Speed [+c]	**Sample > LFO/Spd > SPEED**, 50-200%. Select + INFO for modulation. One Speed voice excludes all active Pitch; multiple Pitch voices can coexist.
Live retry [+c3]	A refused operation stays neutral. At **RETRY**, return its control to neutral, then adjust again; or restart the voice.
Display [+c / +c3]	**Tools > Disp**: Scrn saver and Sig wake [+c]; **Saver gfx** blue-purple-white lines [+c3]. First physical input wakes only.
Tone [+c2 / +c3]	**Pad > Tone**: Fidelity, Drive, Filter 2, Res 2, Slope 1 [+c2]. Phone, ADPCM and u-Law join Fidelity [+c3].
Clip tools [+c2]	**Clip > Pos**: Loop Fade ms, Shift ms, Play Len, Sync 1/32 and Live Flip. Timed Sync + Quant Size required for Play Len / Live Flip.
Fills / grid [+c3]	**Clip > Pos**: Live Flip > Breakcut; Fill every Always/2/4/8/16 bars. **Slicer > Pos > Sync Grid**, with Sync On.
Delay [+c3]	**FX1 > INFO > Type**: Digital / Tape / BBD / Codec. **More** opens Flutter; Main returns.
Reverb [+c2 / +c3]	**FX2 > INFO**: Plate / Room / Hall / Chamber [+c2]. **More** on Room/Hall/Chamber opens seven detailed controls [+c3].
FX3 [+c3]	On Pads, tap FX2 to select it, then tap that cell again: **FX 3**. INFO opens the second independent reverb. Repeat to return to FX2.
Sends / KEY [+c3]	**Pad > Pos > FX1/FX2/FX3 Send**: select + INFO. **KEY** is after CC in the source selector. Maximum 12 routes per pad total.
Return routing [+c3]	**Tools > Rec > FX Out** routes FX1/FX2 returns. **FX3 Out** is on its Plate page or More page for other types.
CV utilities [+c3]	**Tools > Rec > CV In**: shared Scale/Offset, pitch Quantize and Root. Back returns to Rec. Defaults: 100% / 0% / Off / C.
Master [+c2 / +c3]	**Tools > Comp**: compressor controls. **Mstr > Section**: Comp env / Master FX [+c2]. Compressor bus Off/Out1/2/FX1/2/Both [+c3].
MIDI / recording	**Tools > MIDI > Monitor** [+c2]. Record dry while monitoring through pad sends [+c2]; all three sends and their routes retained [+c3].
Reset / counter [+c3]	**Tools > Main > RESET**: choose Tools or FX, then confirm. Double-tap top-left counter: next bar reads 1:1; playback stays in place.

FIRST RELEASE +C / DISPLAY ADDITIONS +C3

Waveform and display

Visual control without changing the recorded sample or its audio level.

Waveform zoom and dial reset [+c]

Press INFO on a loaded pad. Spread two fingers vertically to enlarge amplitude, or pinch to reduce it: **25-3200%**. Horizontal spacing controls time zoom. A diagonal gesture can change both axes. A stationary **two-finger tap** restores 100% amplitude and fits the waveform horizontally.

Double-tap a numeric dial on a pad or an FX parameter grid to restore its base value. Pitch returns to zero and Speed to 100%, for example. **Modulation assignments are retained**, so the effective value can still move.

Screen saver and signal wake [+c]

Tools > Disp > Scrn saver offers disabled or **1, 2, 3, 5, 10, 15, 30 or 60 minutes**. Disabled is the default. Playback continues while the display sleeps. The first touch, HOME/INFO press or encoder movement wakes the display and is consumed; release the control before editing.

Sig wake	Choose **disabled / MIDI / CV trig / both**. A qualifying signal wakes the display and resets its idle timer. That first signal still performs its musical action.
Excluded activity	MIDI Clock and Active Sensing do not wake the screen. CV wake uses rising trigger/gate events; held gates, continuous CV changes and the separate clock input do not wake it.

Optional lines and counter reset [+c3]

Saver gfx > On adds ten single-pixel lines on black, in changing shades from dark blue through purple to white. One line changes every half-second. Off keeps the black blanker. The graphic uses the current brightness and starts only after an enabled saver timeout.

Double-tap the top-left bar:beat counter. A star indicates a pending reset; the next bar is numbered **1:1**. Audio, synchronized clips, clock phase and periodic-fill timing continue unchanged. This display origin is temporary.

Display and signal-wake preferences save globally. The original **MIDI In** tab is named **MIDI**; **Disp** and **Scrn saver** are shortened labels.

SPEED +C / NEUTRAL-THEN-ADJUST RETRY +C3

Speed, Pitch and live retry

Sample parameters > LFO/Spd > SPEED

Pitch-preserving Sample Speed [+c]

Drag SPEED or select it and turn its encoder. **50%** is half speed, **100%** uses native playback, and **200%** is double speed. Select SPEED and press INFO for LFO, external CV and other modulation sources. The effective speed stays within 50-200%.

Pitch active	Multiple native Pitch voices can play together. Any active Pitch operation blocks Speed across the module.
Speed active	One Sample voice owns Speed. It blocks all Pitch operations, including on itself, and prevents a second Speed voice.
Both requested	If a newly starting voice requests both Pitch and Speed, Pitch takes priority. Settings remain editable while an operation is busy.

Release an operation and retry live [+c3]

With Speed modulation depths at zero, return Speed to 100% to release it after its short transition to native playback. Return **all** Pitch operations to neutral to release Pitch, including MIDI pitch bend and note transposition.

A refused voice stays at neutral playback after the owner finishes. It does **not** suddenly change speed or pitch. When the resource becomes available, its label becomes **RETRY**. Restart that voice, or return its blocked control to neutral, let that take effect, then adjust again. If several voices are waiting, none wins automatically.

SPDBUSY / PITBUSY: another operation owns the resource. **RETRY**: it is available, but this voice still needs a deliberate new request. Saved knob values and routes are retained.

An enabled modulation route with nonzero depth reserves its operation through zero crossings. Resetting the base dial does not remove that route. Set its depth to zero to release it. MIDI-note transposition or a draining pitched-grain tail can also keep Pitch occupied.

Speed supports eligible forward Sample playback. Reverse, loop crossfade, active loop-boundary modulation and unsupported source configurations can use native fallback. Speed and its routes save with the pad; older presets default to 100%.

TONE +C2 / CODEC FIDELITY MODES +C3

Tone and Fidelity

Pad parameters > Tone, on Sample, Clip, Slicer and Granular pads.

Fidelity replaces Interp [+c2]

Normal and **HighQ** retain the native choices. **SP1200, SP-12, S950, MPC60 and SP Raw** add sampler-inspired coloration. Fixed-rate choices combine 12-bit quantization with **16, 12, 10, 8, 6, 4 or 2 kHz** rate reduction.

Phone [+c3]	8 kHz mu-law coloration with telephone-band low-pass filtering.
ADPCM [+c3]	IMA-style adaptive 4-bit quantization at 22.05 kHz.
u-Law [+c3]	Mu-law coloration at 16 kHz. The module uses the ASCII label u-Law.

The new modes follow 2k, preserving earlier saved selection numbers. Stereo channels have independent codec histories. Fidelity remains a manual selector.

Drive and two filters [+c2]

Drive	0-100% soft saturation before the filters; default 0%. Select + INFO for modulation.
Slope 1	12 or 24 dB/octave for the original Main-page Filter; default 12 dB. Move Filter away from center to hear the difference. Manual selector.
Filter 2 / Res 2	Independent 24 dB/octave filter. Negative selects low-pass, positive high-pass, center bypasses it. Res 2 defaults to 50%. Both support select + INFO modulation.

The left bank has Fidelity and Drive above Filter 2 and Res 2. Slope 1 sits at the top of the right bank. Tap a bank to give its controls the encoders.

Double-tap resets numeric base values while keeping routes. Returning from modulation restores the same Tone dial and encoder bank. The configuration tab is shortened to **Cnf**. Blank pads expose **LFO/Spd** and Speed at 100%.

CLIP TOOLS +C2 / CLOCKED ADDITIONS +C3

Clip timing, fills and slices

Pad parameters > Pos. Scroll the list to reach the added controls.

Loop Fade [+c2]	0-50 ms, default off. Dips loop edges toward zero to reduce clicks. Capped at a quarter of a short loop; try 2-5 ms initially.
Shift [+c2]	-100 to +100 ms. Positive starts later in the file; negative brings its tail forward. Wraps at the source boundary.
Play Len [+c2]	Full or 1/4-32 bars in quarter-bar steps. Caps at the available source, rounding to the nearest Sync division with a one-division minimum.
Sync 1/32 [+c2]	Adds a finer timed Sync division. Play Len, Live Flip and Fill every require Quant Size enabled and a timed Sync division; not Slice sync or None.

Live Flip [+c2], Breakcut and periodic fills [+c3]

Beat Rpt plays beats 1,2,2,2; **Half Rpt** plays 1,2,1,2; **Rotate** plays 2,3,4,1. **1/8 Rpt** and **1/16 Rpt** repeat subdivisions in beat four. **Breakcut** adds changing sixteenth-note cuts/repeats while keeping the downbeat.

Fill every: Always / 2 / 4 / 8 / 16 bars. At 4 bars, for example, bars 1-3 play normally and bar 4 uses the selected Live Flip. Fills follow the underlying musical clock, including short loops. Counter renumbering does not change their phase. Off rejoins the current source position.

These are playback edits, not a sequencer or another stretch engine. They use no Speed voice. Set Beat Count to match the loop. Cuts through sustained sound can click.

Slice Sync Grid [+c3]

For a Slicer pad with **Sync On**, choose Free / 1 bar / 1/2 / 1/4 / 1/8 / 1/16 / 1/32. Synchronized slice duration rounds to a whole grid unit, minimum one. Free preserves native timing; Sync Off bypasses the grid. **This changes duration, not audio stretching.** It saves with the pad.

Slice Mono fix [+c2]: after switching a polyphonic Sample pad to Slicer, the saved Mono setting is honored. With Trigger, Mono and Play Through On, a new slice cuts the old one even with Quant Size None.

THIRD CUSTOM RELEASE / +C3

Delay and FX return routing

FX1 > INFO for delay. Tools > Rec > FX Out for the first two returns.

Four delay types

Digital	Default; keeps the original delay character.
Tape	Saturation, high/low-frequency loss, slow wow and built-in flutter in the feedback path.
BBD	Darker repeats, light drift and a delay-time-dependent sample hold. Longer delays become coarser.
Codec	16 kHz, 4-bit IMA ADPCM on each trip through the feedback loop. Repeats accumulate codec coloration; no random packet dropout.

The original Delay, Feedback, Cutoff, Width, Beat Sync, Filter and Ping controls remain. Stereo channels have separate histories. Changing Type clears coloration history while retaining the native delay lines and timing. Effect parameters save with the preset.

Flutter

Tap **More** on FX1 to open Flutter, then Main to return. It adds 0-100% flutter depth to any delay type. Default 0 preserves that type's existing character, including Tape's built-in flutter. Numeric double-tap reset applies.

FX Out chooses physical destinations

Out1/2	Delay and FX2 reverb return to the main stereo pair. Default.
Dly FX1/2	Delay returns to the physical FX stereo pair; FX2 reverb stays on main.
Rvb FX1/2	FX2 reverb returns to the FX pair; delay stays on main.
FX1/2	Both returns go to the FX pair.
Dly1 Rvb2	Delay goes to mono FX1, FX2 reverb to mono FX2. Both wet channels are averaged before the mono output.

FX Out saves globally. Moved returns leave the main mix; dry pad output assignments remain. FX3 has its own output setting. Master distortion and Sweep stay on Out1/2. Compressor bus selection is independent.

Reverb and the third send

FX2 > INFO, or select FX3 using the pad-grid toggle.

Reverb types and main controls [+c2]

Plate is the original algorithm and default. **Room, Hall and Chamber** add Size, Diffusion and Width. All types offer Level: 50% is unity, with about -12 to +12 dB available. Decay, Predelay and Damping retain native modulation. Type changes briefly fade and clear the previous tail.

Room / Hall / Chamber > More [+c3]

Bass x / Xover	Bass decay from 0.5x to 2x relative to midrange; center 1x. Crossover approximately 100 Hz-2 kHz.
Mod Rate / Dpth	Internal reverb modulation, approximately 0.1-5 Hz, with adjustable depth.
Early	Early-reflection level.
Hi Cut / Lo Cut	Input high-frequency cutoff approximately 1-14 kHz; low-frequency cutoff approximately 20-500 Hz.
Level / Main	FX2 repeats its output Level here; FX3 uses this square for FX3 Out and keeps Level on its main page. Main returns without clearing the tail.

FX3: a second independent reverb [+c3]

On Pads, tap FX2 once to select it, then tap the selected cell again. Its label becomes **FX 3**. Press INFO for parameters. Another tap on the selected cell returns it to FX2. FX3 has the same four algorithms and detailed controls, with its own settings and tail.

FX3 Out selects Out1/2 / FX1/2 / FX1 / FX2 independently. It is on Plate's main page, or under More for Room/Hall/Chamber. Mono routing averages both wet channels. An enabled compressor processes the destination pair; master distortion and Sweep remain on main only.

FX3 Send is under each pad's **Pos** page and defaults to zero. FX2 and FX3 settings save independently with the preset. The grid toggle changes only which effect is being viewed.

FX3 skips processing while unused and lets a tail run after its last input. Use representative performance loads when checking headroom, especially Speed, two reverbs and compression on Both.

THIRD CUSTOM RELEASE / +C3

Send modulation and KEY

Pad > Pos > FX1 Send / FX2 Send / FX3 Send > select > INFO

Move the sends with modulation

The usual source/amount editor now works for all three pad sends, in every pad mode, including empty pads. LFO, external CV, velocity, KEY, MIDI CC and other native sources can control the send. HOME returns to Pos and the same selected row.

Modulation adds to the manual send level, then clamps to 0-100% before the native send curve. A zero manual send can therefore open under positive modulation. Remove the route to restore the manual level. The original mixer controls set those same underlying levels.

Each send offers three route slots, within the native **12 saved modulation routes per pad total** across all destinations. This limit also includes Pitch, Speed, filters and other assignments.

What happens with polyphony?

A send processes the pad's summed audio. For voice-dependent **LFO, VEL and KEY**, the last active voice in the native list supplies that pad's send modulation; Clip/Slicer uses its last active embedded voice. This is not an independent send envelope for every note and does not promise newest-note priority.

External CV and controller sources are shared independently of that voice selection. When no voice is active, voice-dependent sources contribute zero. Send settings and routes save with the preset; recording retains the send routes too.

KEY exposes native note tracking

Choose **KEY** after CC in the modulation source selector. It uses the engine's existing note-position source, centered on MIDI note 60. Positive or negative amounts can open or close a filter as you move up the keyboard, for example.

KEY is available on supported native destinations and custom Tone, Speed and send destinations. Per-voice parameters retain per-voice tracking; sends use the pad rule above. KEY differs from **PTCH**, the existing pitch-controller source. Older source-selector indices are unchanged.

THIRD CUSTOM RELEASE / +C3

Shared CV utilities

Tools > Rec > CV In. Back returns to the ordinary Rec controls.

Scale and offset external modulation

CV Scale %	0-400%, default 100%.
CV Offset %	-100 to +100%, default 0%. Added after scaling.
Shared scope	Applies to all eight EXT modulation sources together, not separately per socket. IN1/IN2 audio sources are excluded.

At Scale 200% and Offset -50%, a normalized input value of 0.6 becomes 0.7. The transformed modulation value is limited to -2 to +2; each destination keeps its own range limits. Editing these settings also updates a held voltage without needing the CV to move.

Quantize CV-controlled pitch

Quantize offers Off, Major, Minor, Dorian, Phrygian, Lydian, Mixolydian, Harmonic minor, Major pentatonic, Minor pentatonic, Blues and Whole tone. **Root** selects C through B. Defaults are Off and C.

The native CV-pitch source snaps to the nearest allowed note; ties select the lower note. It is **CV-only**: it does not quantize MIDI notes or the final sum of unrelated pitch modulations. Scale/Offset affect the general EXT modulation values; pitch quantizing operates separately on the native CV-pitch source.

Triggers, saving and reset

Raw CV still reaches the native gate/trigger detector first. Trigger thresholds, rising edges and signal wake remain unchanged. A scale or root edit does not synthesize another trigger.

All four settings save globally through the normal Tools store. **RESET > Tools** restores 100% / 0% / Off / C. RESET > FX leaves them alone. Leaving Rec closes the subpage, so returning to Rec shows its ordinary controls.

For an initial check, keep Scale 100%, Offset 0% and Quantize Off, then change one setting at a time. The default settings preserve native CV behavior.

MASTER CONTROLS +C2 / ROUTING AND RESET +C3

Master processing and RESET

Tools > Comp and Tools > Mstr > Section

Compressor controls [+c2]

Comp page	Threshold -48 to 0 dB; Ratio 1:1 to 20:1; Gain -12 to +24 dB; parallel Mix 0% dry to 100% processed.
Comp env	Tools > Mstr > Section > Comp env: Attack 1-200 ms and Release 10-2000 ms. Defaults: 10 ms and 250 ms.
Defaults	Threshold -4 dB, Ratio 4:1, Gain -4 dB and Mix 100%. The original compressor algorithm remains in use.

Compressor bus [+c3]

Off / Out1/2 / FX1/2 / Both appears under Tools > Main and Mstr > Comp env. These are two views of **the same setting**. FX1/2 means the physical FX output pair. Both uses independent histories/detectors with shared controls. Mono assignments join their output pair before compression.

Master FX [+c2]

Mstr > Section > Master FX offers **Soft, Hard, Fold or Tube** distortion, or Off; Drive adds 0 to +24 dB. Sweep left for low-pass, right for high-pass, center for dry. On Out1/2 the order is distortion, Sweep, then optional compression. These settings save globally; changing compressor buses does not move distortion or Sweep.

RESET with two scopes [+c3]

Tap **Tools > Main > RESET**, beside the version. Choose Tools or FX, then **Reset** to confirm. Cancel or leave the panel to dismiss it.

Tools	Resets native/custom global settings: display, CV, MIDI/clock, recording options, routing, compressor and master processing. Saves one global-settings snapshot.
FX	Resets FX1, FX2 and FX3 scalar settings. Changes the current preset; save the preset to retain the reset. Preserves Tools values.

Both scopes preserve pads, samples, pad parameters, send levels and modulation routes. Neither deletes files or overwrites a saved preset.

RECORDING AND MIDI MONITOR +C2 / SEND EXPANSION +C3

Recording and MIDI monitor

Monitor a dry take through effects, or inspect MIDI arriving at the module.

Recording keeps the sends

[+c2] FX1/FX2 send levels survive recording, and the selected pad's sends process live input monitoring. **[+c3]** This includes FX3 and the modulation routes on all three sends. External CV/controller modulation can work while the pad has no active voice.

Monitor On / Auto / Off keeps its native meaning. Selecting another pad changes which send settings monitor uses. The WAV contains the input, including input gain; the effects heard through sends stay out of the recorded sample.

Tools > MIDI > Monitor [+c2]

Tap Monitor beside the version number. Capture starts on entry. Seven visible rows show the newest entries from a 32-event history. Each row gives elapsed time, input group, channel and message. **0.350 A 03 CC 74=96**, for example, means CC 74 at value 96 on channel 3 after 350 ms.

Freeze / Resume	Freeze holds the displayed history while capture and playback continue. Encoder 2 scrolls frozen rows. Resume returns to the latest events.
Channel filter	Touch Ch All / Ch 1-16 or use encoder 1. Transport messages appear with every channel setting.
Clear / Back	Clear restarts elapsed time and clears the log. Back stops capture and returns to MIDI settings.
CLK / SENSE	Separate activity lights for MIDI Clock and Active Sensing. Rows show notes, CC, program changes, pressure, pitch bend, transport and reset messages.

A/B are native input groups. B includes the USB queue and one parser path, so a group is not a unique jack identifier. SysEx and system-common messages are discarded before this monitor. Signal Wake is configured separately under Disp.

Internal improvements [+c3]

The memory refactor moves the dial-reset workspace into reserved SDRAM and compacts the voice ledger, Speed routing and status caches. This makes room for additions while retaining existing operations and the Pitch/Speed limit.

THIRD CUSTOM RELEASE / 2.3.4+C3

Install, recover and credits

Bitbox MkII only. The patcher always takes official firmware 2.3.4 as input.

Patch locally and install

1. **Back up the entire microSD card**, including projects, samples and presets. Download official 2.3.4 and extract the ZIP. At the browser patcher, select the original BITBOXMK2.bin, even when upgrading from +c or +c2. Patching runs in your browser; the firmware is not uploaded. Input and output SHA-256 checks guard the patch.

2. Put the patched file in the card root, named exactly **BITBOXMK2.bin**. Remove duplicate-download suffixes such as (1). Power off, insert the card, then hold **INFO** while powering on. Release INFO when the update message appears.

3. Allow about 30 seconds. Keep power on and wait for automatic reboot. Check **2.3.4+c3** at Menu's lower-right or Tools' top-right. See the official MkII manual for the complete flashing procedure.

Official input: 674,896 bytes. SHA-256:
35ab7995ad0eca8484f3b28bc76389f10b2037dbb21a4a394e950d9329073e86

Customized +c3 output (748,612 bytes). SHA-256:
89b993d89db865a8a5dfd46f64ca337edff6cde66e1f2620e42bc984639d747b

Return to official firmware

Replace the patched file on the card with the original unmodified **BITBOXMK2.bin** from the official ZIP. Repeat the INFO-held power-on update and wait for reboot. Confirm **2.3.4** in Menu. Keep backups of presets made before upgrading.

Reference and acknowledgment

First release (+c) video overview. The video demonstrates the original features; later-release controls are covered in this guide.

Original firmware and hardware: **1010music**. Custom integration and patcher: **vwal**. Adapted code and groundwork: semierendonmez-commits / bitbox-mk2-semi; Portkit contributors; Justin Joe / blackbox-mod for Fidelity/filter work and reverse engineering; **Emilie Gillet** for stmlib filtering. Applicable MIT notices accompany the patch source. They do not license 1010music's original firmware.

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